

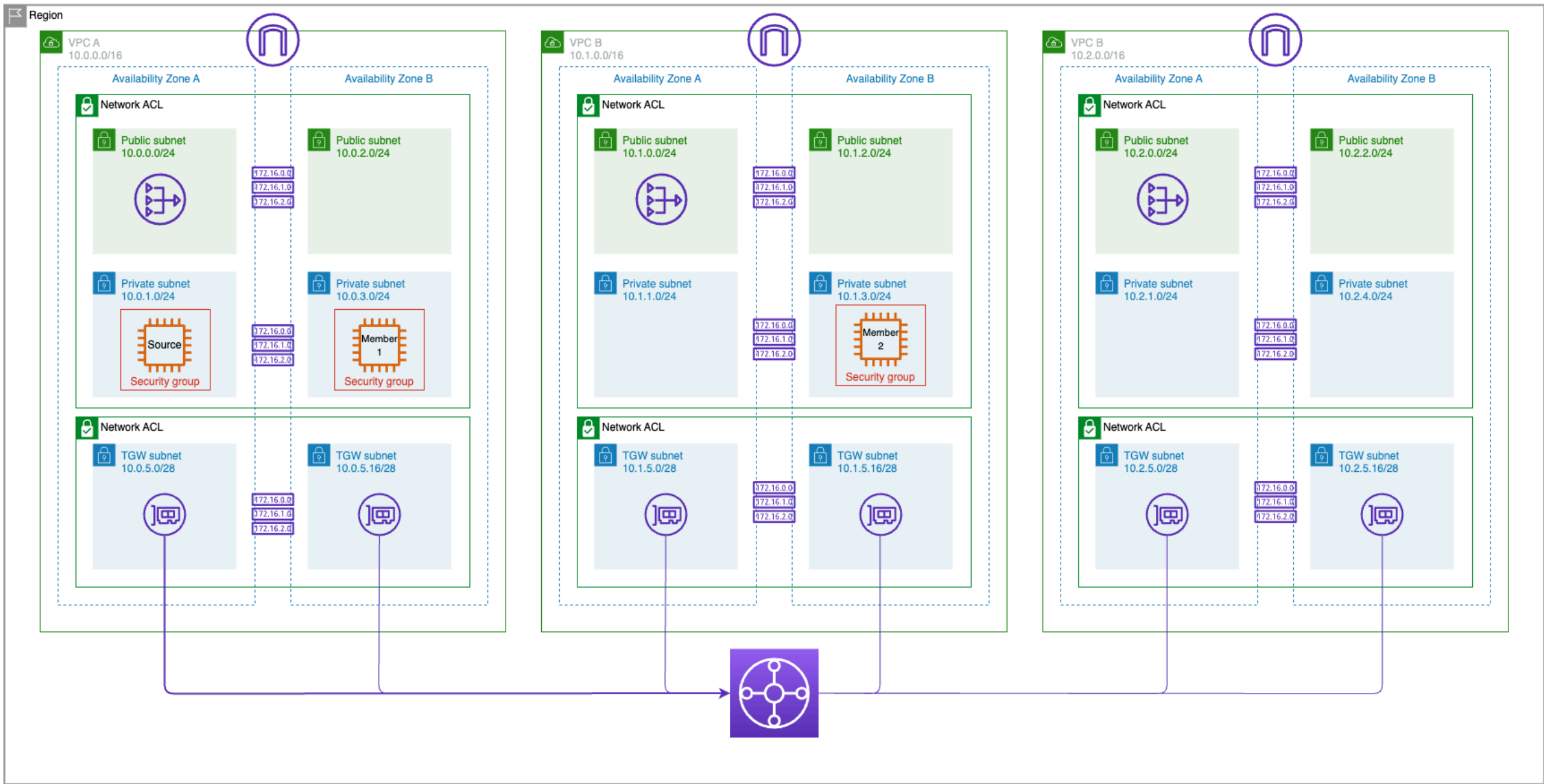
Lab 1: Multicast in AWS

Review Multicast Networking

Multicast is a communication protocol used for delivering a single stream of data to multiple receiving computers simultaneously. AWS Transit Gateway supports routing multicast traffic between hosts in the same subnet or between subnets of attached VPCs, and it serves as a multicast router for instances sending traffic destined for multiple receiving instances. While an AWS VPC by itself does not support multicast, Transit Gateway can provide this new capability. Transit Gateway will act as the rendezvous point, receiving the packets from the multicast source, replicate it, and send it to the multicast receiver. Transit Gateway supports routing multicast using the Internet Group Management Protocol (IGMP) protocol or static source and member configurations.

Understanding the Lab Environment

In this lab, you will configure a multicast domain on the Transit Gateway created in [Foundational Topics - Multiple VPCs](#). Recall at the end of that lab, you provisioned 3 VPCs and connected them using Transit Gateway with unicast networking. You had enabled multicast routing support when you created the Transit Gateway. The following diagram is the network topology for this lab.



For this lab, please be aware of the following:

- You will create a transit gateway multicast domain on the existing Transit Gateway.
- Multicast group membership is managed using the Amazon VPC Console, the AWS CLI, or IGMP.
 - In this lab, IGMPv2 is automatically configured on the EC2 instances created by the CloudFormation stack. IGMPv3 is currently not supported.
 - To check the IGMP version run `cat /proc/net/igmp` on the EC2 instances.
- A subnet can only be in one multicast domain.

For additional considerations outside of those needed for this lab, please see the [Multicast on Transit Gateways](#) documentation.

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AWS Networking Workshop

Foundational Topics

Advanced Topics

AWS Gateway Load Balancer

AWS Transit Gateway Multicast

Prerequisites

Lab 1: Multicast in AWS

1. Deploy Lab Resources

2. Set Up Transit Gateway

3. Explore Multicast

4. Review & Clean Up

Contributors & Revision History

Workshop catalog in AWS Builder Center

Content preferences

Language

English

1. Deploy Lab Resources

If you did not complete Foundational Topics - Multiple VPCs, you must complete the Prerequisites section.

Create the CloudFormation Stack

A CloudFormation template has been provided to create the EC2, and Clouwatch dashboard resources used in this lab. The CloudFormation template also modifies the VPC A and VPC B inbound security groups to allow IGMP and the multicast test traffic.

Type	Protocol	Port range	Source	Description
Custom Protocol	IGMP(2)	All	0.0.0.0/32	IGMP query
Custom UDP Protocol	UDP	8123	10.0.0.0/8	Inbound multicast traffic

CloudFormation template

- Open the AWS CloudFormation console <https://console.aws.amazon.com/cloudformation/>.
- Click **Create stack** (With new resources).
- Choose **Template is ready**, then choose **Upload a template file**, and click **Choose file**.
- Select the file `TGW-Multicast-Lab1.yaml`, click **Open**, and then click **Next**.

Prerequisite - Prepare template

Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

Template is ready

Use a sample template

Create template in Designer

Specify template

A template is a JSON or YAML file that describes your stack's resources and properties.

Template source

Selecting a template generates an Amazon S3 URL where it will be stored.

Amazon S3 URL

Upload a template file

Upload a template file

Choose file

TGW-Multicast-Lab1.yaml

JSON or YAML formatted file

S3 URL:

https://s3-external-1.amazonaws.com/cf-templates-

-us-east-1/2022097eMU-TGW-Multicast-Lab1.ya

ml

View in Designer

Cancel

Next

- Name the stack `NetworkingWorkshop-Multicast-Lab1`. This stack will create three (3) EC2 instances, and a CloudWatch dashboard. The EC2 Instances automatically update, install required network tools for the exploratory section of this lab, and reboot after launch. The status check for each EC2 instance should display *2/2 checks passed* when complete.
- In the **EC2 Setup** section, select a subnet and corresponding security group for the source and member EC2 instances. The EC2 instances can be in the same or different VPCs and subnets, or any combination thereof. You will connect to the EC2 instances using Session Manager.

- Source Subnet: VPC A Private Subnet AZ1
- Source Security Group: VPC A Security Group
- Member Subnet: VPC A Private Subnet AZ2
- Member Security Group: VPC A Security Group
- Member Subnet: VPC B Private Subnet AZ2
- Member Security Group: VPC B Security Group
- EC2 Instance Type: Leave as default
- Latest Amazon Linux 2 AMI: Leave as default

Stack name

ImmersionDay-Multicast-Lab1

Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

Multicast Setup

Group IP

Enter an IPv4 multicast address in the range 239.0.0.0 to 239.255.255.255.

239.0.0.100

EC2 Setup

Source Subnet

Select the subnet where multicast traffic will originate.

subnet-00aed8ba70f7c83ba

VPC A Private Subnet AZ1

Source Security Group

sg-02884d881fcc31e22

VPC A Security Group

Member Subnet

Select the first subnet where multicast traffic will be sent.

subnet-0a6d94c9ef2092161

VPC A Private Subnet AZ2

Member Security Group

sg-02884d881fcc31e22

VPC A Security Group

Member Subnet

Select the second subnet where multicast traffic will be sent.

subnet-011ef2ef8b37841fc

VPC B Private Subnet AZ1

Member Security Group

sg-07ddec89523672519

VPC B Security Group

EC2 Instance Type

Select the instance type for the source and member instances.

t2.micro

/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2

Cancel

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- Click **Next**, leave all options default, and click **Next** again.
- Scroll to the bottom of the page and click **Submit**.

If you receive the error You have specified two resources that belong to different networks during the stack creation, your Security Group to VPC mapping is incorrect.

- Wait for the stack to be created and then click the **Outputs** tab. Take note of the IPv4 Address for each member instance.

ImmersionDay-Multicast-Lab1

Delete

Update

Stack actions

Create stack

Stack info

Events

Resources

Outputs

Parameters

Template

Change sets

Outputs (2)

Search outputs

< 1 >

Key

Value

Description

Export name

MulticastMember1

10.0.3.170

Multicast-Member-1 IPv4 Address

-

MulticastMember2

10.1.3.103

Multicast-Member-2 IPv4 Address

-

EC2 Instance

IPv4 Address

MulticastMember1

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2. Set Up Transit Gateway

Setting Up Multicast on Transit Gateway

You may use the existing Transit Gateway created in the foundational workshop since multicast was enabled at creation time.

Create a Static Multicast Domain

A multicast domain allows segmentation of a multicast network into different domains, and makes the Transit Gateway act as multiple multicast routers. You define multicast domain membership at the subnet level. For this domain, you will configure static sources support which allows the addition of multicast members as sources. Only multicast sources are allowed to send multicast data.

1. In the left hand pane of the VPC Dashboard scroll down and click on [Transit Gateway Multicast](#), and click **Create transit gateway multicast domain**.
2. Name the domain `TGW-Multicast-Static` and choose the multicast Transit Gateway (TGW) from the *Transit gateway ID* dropdown.
3. Select **Static sources support** and click **Create transit gateway multicast domain**.

Create transit gateway multicast domain

Info

A multicast domain controls how traffic flows for all associated subnets. A multicast domain can only be created in transit gateway with the multicast support option enabled.

Details

Name tag - optional

Creates a tag with the key set to Name and the value set to the specified string.

TGW-Multicast-Static

Transit gateway ID

Info

tgw- (TGW)

Configure the transit gateway multicast domain

☐ IGMPv2 support

Info

☒ Static sources support

Info

☐ Auto accept shared associations

Info

4. Select the static multicast domain (TGW-Multicast-Static), click the **Associations** tab, and click **Create association**.
5. For each VPC, choose the two private subnets that will be allowed to send and receive multicast traffic, and click **Create association**.

Create association

Info

Associating a subnet to a multicast domain enables the subnet to send and receive multicast traffic. A subnet can only be associated to one multicast domain.

Details

Transit gateway ID

tgw-05ef1d8d660a89017

Transit gateway multicast domain ID

tgw-mcast-domain-0c6d89838f112ba1f

Choose attachment to associate

tgw-attach-08a5ebb8cf6d7abab (VPC A Attachment)

Choose subnets to associate

Info

Select subnets

subnet-00aed8ba70f7c83ba (VPC A Private Subnet AZ1)

us-east-1a vpc-0e5ad3b483cd049e0

subnet-0a6d94c9ef2092161 (VPC A Private Subnet AZ2)

us-east-1b vpc-0e5ad3b483cd049e0

Cancel

Create association

Make sure you repeat the previous step for VPC B and VPC C

6. Select the static multicast domain (TGW-Multicast-Static), click the **Groups** tab, and click **Add source**. Enter `239.0.0.100` for the *Group IP address*, select the source network interface, and click **Add group sources**.

Add group sources

Info

Adding a source to a multicast group enables the network interface to send multicast traffic to members of the multicast group.

Details

Transit gateway ID

tgw-

Transit gateway multicast domain ID

tgw-mcast-domain-

Group IP address

Requires a valid IPv4 or IPv6 IP Address in the 224.0.0.0/4 or ff00::/8 CIDR range.

239.0.0.100

Available network interfaces

Info

Refresh

Create network interface

Filter network interfaces

< 1 2 >

☒

Name

Network Interface ID

Subnet ID

Availability Zone

☒

Multicast-Source

eni-

subnet-

us-east-1a

7. Click the **Groups** tab and click **Add member**. Enter `239.0.0.100` for the *Group IP address*, select the member network interfaces, and click **Add Group Members**.

Add group members

Info

Adding a member to a multicast group enables the network interface to receive multicast traffic sent by the sources of the

Details

Transit gateway ID

tgw-

Transit gateway multicast domain ID

tgw-mcast-domain-

Group IP address

Requires a valid IPv4 or IPv6 IP Address in the 224.0.0.0/4 or ff00::/8 CIDR range.

239.0.0.100

Available network interfaces

Info

Refresh

Create network interface

Filter network interfaces

< 1 2 >

☒

Name

Network Interface ID

Subnet ID

Availability Zone

☒

Multicast-Member-1

eni-

subnet-

us-east-1b

☒

Multicast-Member-2

eni-

subnet-

us-east-1b

☐

-

eni-

subnet-

us-east-1a

☐

-

eni-

subnet-

us-east-1a

☐

Multicast-Source

eni-

subnet-

us-east-1a

Create a Dynamic Multicast Domain

Transit Gateway supports Internet Group Management Protocol (IGMP) for simplified deployment and management of multicast applications. For this domain, you will configure IGMP support which enables Transit Gateway to dynamically add and delete multicast members based on IGMP protocol interactions.

This feature provides real-time visibility into the multicast network and enables you to accurately keep track of group membership changes over time. You cannot use IGMPv2 and Static Sources support at the same time.

1. In the navigation pane, choose **Transit Gateway Multicast**, and click **Create transit gateway multicast domain**.
2. Name the domain `TGW-Multicast-Dynamic` and choose the multicast Transit Gateway (TGW) from the *Transit gateway ID* dropdown.
3. Select **IGMPv2 support** and click **Create transit gateway multicast domain**.

Create transit gateway multicast domain

Info

A multicast domain controls how traffic flows for all associated subnets. A multicast domain can only be created in transit gateway with the multicast support option enabled.

Details

Name tag - optional

Creates a tag with the key set to Name and the value set to the specified string.

TGW-Multicast-Dynamic

Transit gateway ID

Info

tgw- (TGW)

Configure the transit gateway multicast domain

☒ IGMPv2 support

Info

☐ Static sources support

Info

☐ Auto accept shared associations

Info

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3. Explore Multicast

Connecting to the Application Instances

Session Manager provides secure and auditable instance management without the need to open inbound ports, maintain bastion hosts, or manage SSH keys. Follow the steps below to connect to the application instances using Session Manager.

1. Open the **Amazon EC2** console at <https://console.aws.amazon.com/ec2/>.
2. In the navigation pane, choose **Instances**.
3. Select the source instance (Multicast-Source) and choose **Connect**.
4. Click the **Session Manager** tab and choose **Connect**.

🔔 If you see the warning **Can't connect to your instance**, click the **SSH client** tab and then the **Session Manager** tab to refresh.

Connect to instance

Info

Connect to your instance i-06e08317681ef0b50 (Multicast-Member-1) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

⚠ Can't connect to your instance

To connect to an instance using Session Manager, Systems Manager requires an IAM instance profile. You can create an instance profile and assign it to your instance by using a Systems Manager Quick Setup [host management configuration](#).

If you still can't connect to your instance, or if you receive an error, including an error about SSM Agent, see [Troubleshooting Session Manager](#).

Open Systems Manager Quick Setup

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

CancelConnect

5. Repeat these steps to connect to the member instances (Multicast-Member-1, Multicast-Member-2).

EC2 > Instances > i-

Connect to instance

Connect to instance

Info

Connect to your instance i- (Multicast-Source) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 Serial Console

Session Manager usage:

- Connect to your instance without SSH keys or a bastion host.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

CancelConnect

Running a Multicast Application

As part of the CloudFormation stack, the EC2 instances are provisioned with a Python-based script ([mcast.py](#)) that will be used to send and receive multicast traffic. This script is the multicast demo from the Python Software Foundation's CPython distribution; it is bundled with this workshop (see the attribution header in the file) rather than downloaded from an external host. It is renamed to `mcast_app` and automatically configured to join the 239.0.0.100 group IP address.

Error

If you get an error message trying to open `mcast_app` such as `command not found` or `Permission denied`, or the app closes immediately, run the command `sudo mcast_app_install.sh` to force a re-install.

1. On the source instance, run the command `mcast_app -s`.
2. On the member instances, run the command `mcast_app`.
3. Observe that the member instances begin receiving multicast traffic from the source instance. The output will contain the source IPv4 address, source port, and timestamp.

```
sh-4.2$ mcast_app
('10.0.0.13', 55942) '1649392726.889383'
('10.0.0.13', 55942) '1649392727.890499'
('10.0.0.13', 55942) '1649392728.891612'
('10.0.0.13', 55942) '1649392729.892724'
('10.0.0.13', 55942) '1649392730.893835'
('10.0.0.13', 55942) '1649392731.894951'
('10.0.0.13', 55942) '1649392732.896127'
('10.0.0.13', 55942) '1649392733.897244'
('10.0.0.13', 55942) '1649392734.898422'
('10.0.0.13', 55942) '1649392735.899538'
```

4. Press **CTRL+C** on the member instances to stop the multicast application.
5. While still on the member instances, run the command `sudo tcpdump -i eth0 -n port 8123`.

```
sh-4.2$ sudo tcpdump -i eth0 -n port 8123
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
04:40:04.976169 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:06.976444 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:07.979622 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:08.980732 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 17
04:40:09.981811 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:10.982925 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:11.983980 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:12.985189 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
04:40:13.986306 IP 10.0.0.13.55942 > 239.0.0.100.8123: UDP, length 18
```

6. Observe that the member instances continue to receive multicast traffic. *Why is this the case?*

► Check Your Understanding

7. Press **CTRL+C** on the source and member instances to stop the multicast application and packet capture.

Using IGMP for Group Membership

IGMP support on Transit Gateway makes it easier to scale up multicast workloads, while also simplifying the management of multicast group membership and network deployment. You do not need to configure static multicast groups, sources, and receivers when building a multicast network in AWS. Transit Gateway dynamically adds and deletes multicast members based on IGMP protocol interactions.

Configure the Multicast Domain

1. Open the **Amazon VPC** console <https://console.aws.amazon.com/vpc/>.
2. In the navigation pane, choose **Transit Gateway Multicast**.
3. Disassociate all of the subnets from the static multicast domain (TGW-Multicast-Static).
4. Reassociate the subnets to the dynamic multicast domain (TGW-Multicast-Dynamic).

Run the Multicast Application

1. On the source instance, run the command `mcast_app -s`.
2. On the member instances, run the command `mcast_app`.
3. Navigate to the **TGW-Multicast-Dynamic** multicast domain and select Groups.
4. Observe that the 239.0.0.100 multicast group was automatically created.

Groups (4)

🔍 Filter groups

☐	Group IP address	Network interface ID	Member type
☐	224.0.0.1	eni-	Igmp
☐	224.0.0.1	eni-	Igmp
☐	239.0.0.100	eni-	Igmp
☐	239.0.0.100	eni-	Igmp

🔔 Transit Gateway automatically joins the 224.0.0.1 (All Systems) multicast group and sends membership query packets to all the IGMP members so that it can track multicast group membership.

5. After a few moments, confirm that the member instances are receiving multicast traffic from the source instance.

```
sh-4.2$ mcast_app
('10.0.0.13', 42953) '1649393310.636771'
('10.0.0.13', 42953) '1649393311.639938'
('10.0.0.13', 42953) '1649393312.639045'
('10.0.0.13', 42953) '1649393313.640239'
('10.0.0.13', 42953) '1649393314.641349'
('10.0.0.13', 42953) '1649393315.642443'
('10.0.0.13', 42953) '1649393316.64356'
('10.0.0.13', 42953) '1649393317.644687'
('10.0.0.13', 42953) '1649393318.645884'
('10.0.0.13', 42953) '1649393319.646721'
```

6. Press **CTRL+C** on one of the member instances to stop the multicast application.
7. On the same member instance, run the command `sudo tcpdump -i eth0 -n port 8123`.
8. Observe that this member instance is no longer receiving multicast traffic; however, the other member instance is still receiving multicast traffic. *Why is this the case?*

► Check Your Understanding

9. Press **CTRL+C** on the source and member instances to stop the multicast application and packet capture.

Testing Bandwidth Conservation

As part of the CloudFormation stack, the EC2 instances download and install iPerf, an open-source bandwidth measurement tool. You will use iPerf to confirm that multicast conserves bandwidth by delivering a single stream of data to multiple receiving computers simultaneously.

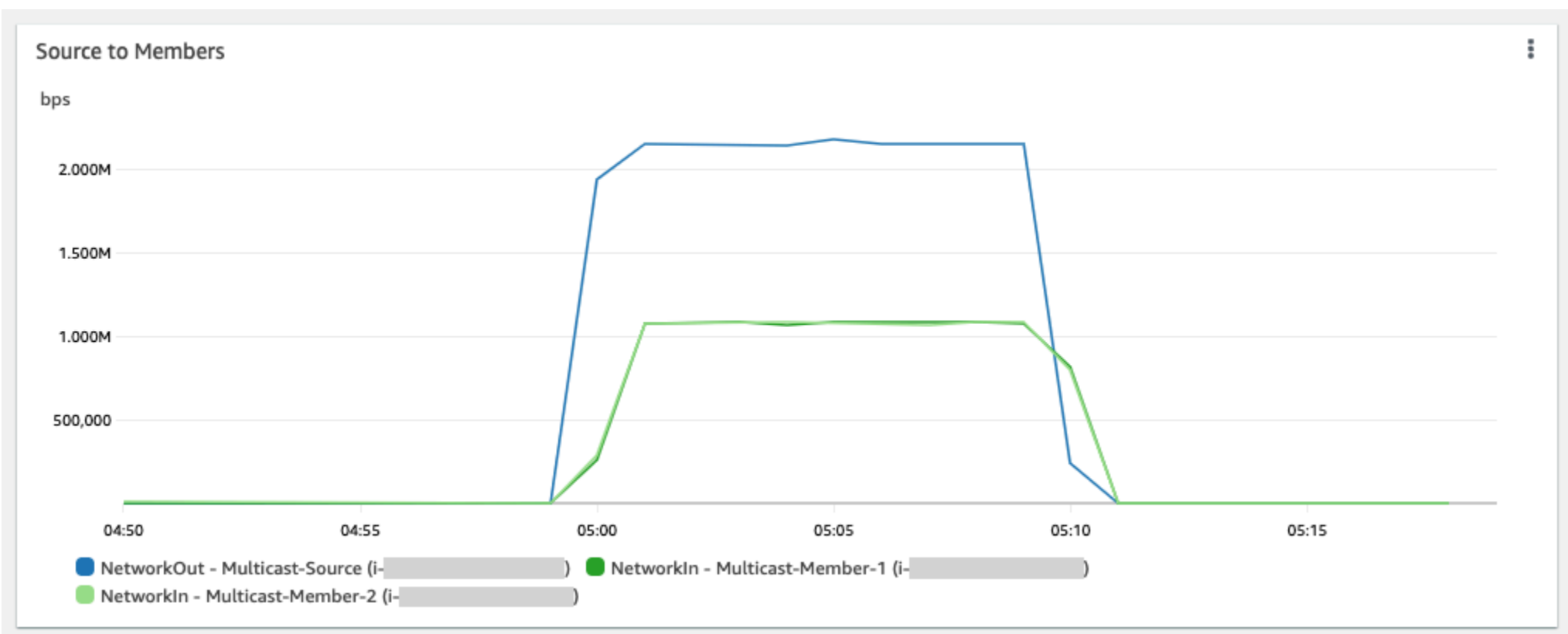
Unicast

1. On the member instances, run the command `iperf -s -p 8123 -u`.
2. On the source instance, run the following commands. Replace **<MEMBER_1_IP>** and **<MEMBER_2_IP>** with the IPv4 Addresses that you noted in step 9 of the [Deploy Lab Resources](#) section of this lab.

```
iperf -c <MEMBER_1_IP> -p 8123 -u -b 1M -t 600 &
iperf -c <MEMBER_2_IP> -p 8123 -u -b 1M -t 600 &
```

3. Open the Amazon CloudWatch console at <https://console.aws.amazon.com/cloudwatch/>.
4. In the navigation pane, choose **Dashboards**.
5. Choose the network dashboard (Network-Throughput).
6. In the refresh dropdown, choose **10s** and allow 2 to 3 minutes for the graph to populate.
7. Observe that **NetworkOut** on the source instance is at twice the level as **NetworkIn** on the member instances. *Why is this the case?*

► Check Your Understanding

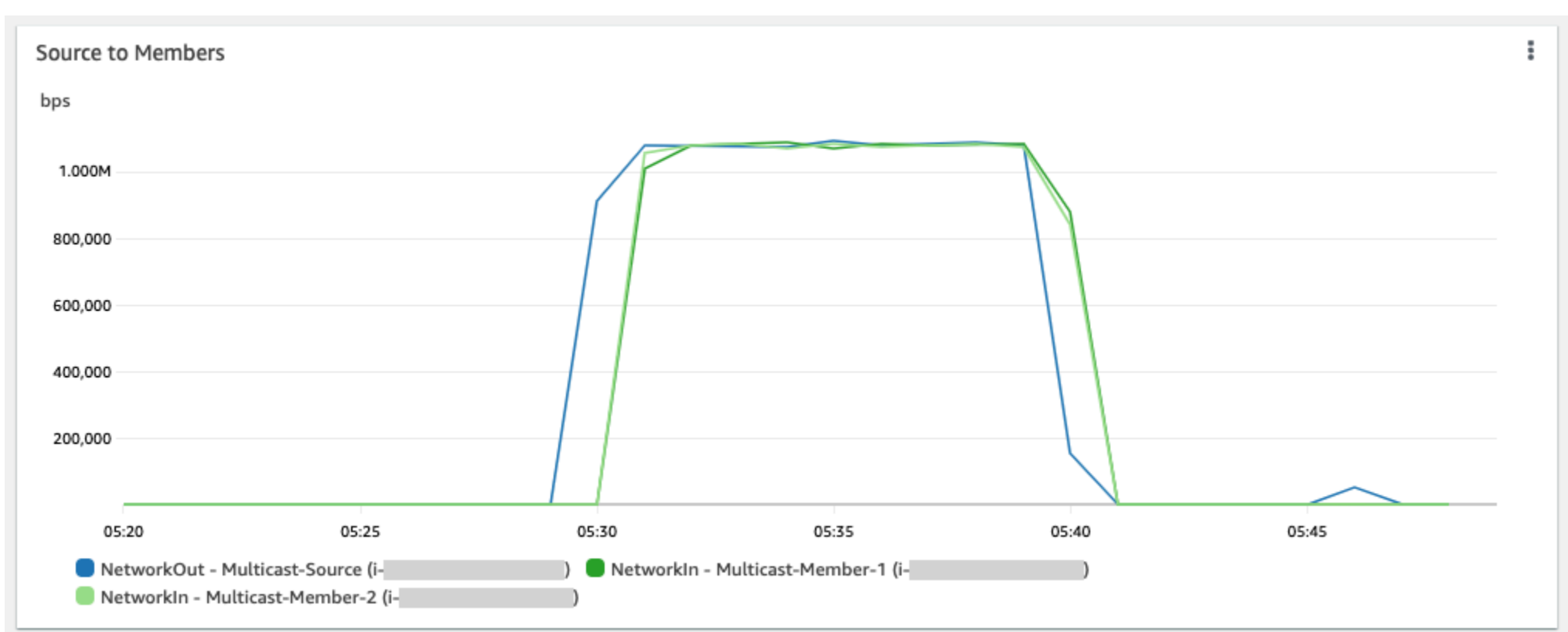


8. Press **CTRL+C** on the source and member instances to stop the unicast throughput test.

Multicast

1. On the member instances, run the following command: `iperf -s -B 239.0.0.100 -p 8123 -u`
2. On the source instance, run the following command: `iperf -c 239.0.0.100 -p 8123 -u -b 1M -T 32 -t 600`
3. Navigate back to the network dashboard (Network-Throughput) and allow 2 to 3 minutes for the graph to populate.
4. Observe that **NetworkOut** on the source instance is at the same level as **NetworkIn** on both the member instances. *Why is this the case?*

► Check Your Understanding



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4. Review & Clean Up

Review

In this lab, you deployed a separate Transit Gateway with multicast support for static and dynamic domains. Next, you verified multicast connectivity across multiple VPCs and subnets using a sample Python script. You performed a packet capture to understand how multicast traffic is routed differently in a static versus dynamic domains. Lastly, using iPerf and CloudWatch you confirmed that Transit Gateway delivered a single stream of data to multiple receiving instances simultaneously.

Clean Up

If you are using your own AWS Account to conduct this workshop and you are finished, complete the follow steps to clean up.

 Make sure you terminate / delete the resources below to avoid unnecessary charges.

Disassociate the subnets from the multicast domain.

- **Step-by-Step Instructions**

Delete each multicast domain.

- **Step-by-Step Instructions**

Delete the CloudFormation stack.

- **Step-by-Step Instructions**

Please make a selection:

- **Option 1: You started this lab at the start**
- **Option 2: You started the lab in the Multi VPCs section by deploying the CloudFormation template**
- **Option 3: You started the lab in the Basic Security or Multicast sections by deploying the CloudFormation template**

Finished

Once you have completed the above steps, you are finished. If you want to continue the workshop later, you can skip this step and use the CloudFormation template in the next step to build the prerequisites.

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Accept

Decline

Customize